

National University of Computer and Emerging Sciences



Lab Manual 09

Programming Fundamentals

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Activities:

Note: Include input validation in programs wherever applicable. Don't use any build-in function.

1. *Unique Element Finder*

Write a program that takes an array of N integers and finds the unique elements that are not repeated in the array. If no unique elements are found, display a message stating that.

2. *Array Shift Operations*

Create a program that takes an array of N integers and allows the user to perform left or right circular shifts. After each shift operation, display the updated array. The user specifies the direction (left or right) and the number of shifts.

Input: [1,2,3,4,5] number of shift = 4 direction = 'r'

Output: [2,3,4,5,1]

3. *Majority Element*

Given an array of N integers, find the element that occurs more than $N/2$ times in the array (the majority element). If no such element exists, output a message stating that.

4. *Rearrange Even and Odd Numbers*

Write a program that takes an array of N integers and rearranges the array such that all even numbers appear before all odd numbers, maintaining their relative order.

Example Input:

2, 7, 3, 8, 1, 4, 15, 20, 16

Example Output:

2, 8, 4, 20, 16, 7, 3, 1, 15

5. *Sum of Digits of Array Elements*

Write a program that takes an array of N integers and calculates the sum of digits for each element. Store the sum of digits in another array and display both the original and the resulting arrays.

Input Array:

12, 17, 3, 38, 101, 64, 15, 20, 16

Output Array:

3, 8, 3, 11, 2, 10, 6, 2, 7

6. *Array Intersection*

Write a program that takes two arrays of integers and finds their intersection (common elements). The result should include only the unique common elements.

7. *Vowels and Consonants*

Write a program that takes a character array (string) and counts the number of vowels and consonants in it. The program should ignore case and non-alphabetic characters.

Example Input:

Enter a string: Hello, World!

Example Output:

Vowels: 3, Consonants: 7

8. *Palindrome*

Write a program that checks if a given character array is a palindrome. A palindrome is a word that reads the same forward and backward (ignoring case sensitivity and non-alphabetic characters).

Example Input:

Input string: "A man, a plan, a canal, Panama!"

Example Output:

The input string is a palindrome.

Example Input:

Input string: "Hello, World!"

Example Output:

The input string is not a palindrome.

9. *Reverse Order*

Write a program that takes a character array (sentence) as input and reverses the order of the words in the sentence, while keeping the characters within each word intact.

Example Input:

Enter a sentence: This is a test

Example Output:

test a is This